THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

COURSE CODE: ICT 302

COURSE TITLE: COMPUTER PACKAGES

PROGRAM: HND 1 ELECTRICAL ELECTRONIC ENGINEERING.

CHAPTER ONE

Database Management Package

1.1 Definition and concept of Database

A database is an organized collection of information (data) for easy storage, manipulation and fast retrieval. Examples of databases are all around us. Phone books, dictionaries, encyclopedias, stock quotes in the newspapers, WAEC results and admission lists are all databases. The data may be in form text, numbers or encoded graphics. Drug store owners have use databases for years by keeping lists of drugs thy sell on cards and classroom teachers develop database of information of students assigned to their class. Simply stated, if you have a collection of information that is arranged in such a way that the information can be easily manipulated and retrieved, the you have database.

1.2 Ways of keeping database

There are two ways of keeping database: -

i. Manual Filing – these are paper documents (i.e. list, index cards and file folders) kept in fil cabinets trays.
ii. Electronic Filing – these are document/files saved in a computer on the hard drive, Zip drive DVD, CD-ROM etc.

1.3 Advantages of Electronic Database over Manual Filing

i. Takes up less storage space (save filing space)
ii. No paper records to damage
iii. Information can be located quickly (using search/query facility)
iv. Information can be sorted into any order quickly using the sort facility
v. Information can be added/deleted/amended quickly and easily
vi. A query can be used to create a report
vii. Multi-user access
viii. Remote access
ix. Provide necessary security to protect data stored (security passwords restrict access to only authorized person)
x. Backups can be easily created

Before computers, database information was kept manually in the form of lists on index cards, in file folders and so on. In many cases, databases are still stored in this manner. A computerized database is like an electronic filing cabinet of information arranged for easy access or for a specific purpose. Working with non-electronic database is usually tedious because it can involve a great deal of paper or card shuffling to derive any summary information or reports. The computer can do the same work much quicker and with less effort.

By using programs developed specifically for storing and manipulating information in the form of a database, a user may access a particular item, manipulate the database into another form, or even isolate parts of database for special attention. A summary report may be output based on the contents of the database. The programs that allow a user to work with an electronic database are called ***Database Management Packages*** and have built-in functions that make this work easy.

All database management packages can perform the following three basic operations: -

i. Sorting and retrieving information
ii. Manipulating and rearranging information
iii. Generating reports

Database management packages can be broadly divided into groups: -

i. File Processing System
ii. Database Management System

The primary difference between the two packages is the number of files the package can manage. A File Processing System (FPS) can work with only one file at a time, while a Database Management System (DBMS) has the capability to work simultaneously with multiple files.

A FPS can be used for working with many types of lists, including mailing lists, membership rolls, customer lists, parts lists, and so on, to carry out the three basic operations listed above. However, whenever needed information appears on different lists in separate files, a DBMS must be used to work with multiple files.

1.4 Examples of Database Systems

The following are examples of database applications: -

- Banking
- Automated Teller Machines
- Universities
- Credit card transaction

- Finance
- Computerized library systems
- Flight reservation systems
- Telecommunication
- Sales
- Manufacturing
- Human resources

i. Automated Teller Machines: - When you step up to an automated teller machine (ATM)and insert your card, information about you is immediately looked up in the bank's database. Any transaction you make is instantly recorded.

ii. Flight reservation system: - When you make a reservation with an airline, your name and address are entered into its database, and a seat on a specific flight is reserve for you in the same database. Should you call later and give your name, the airline can locate the seat that was assigned by searching the database.

iii. Credit card transactions: - When you make a credit card charge, your card is scanned, a computer is dialed up, and information about you is looked up and new information recorded

iv. Sales (e.g. mail-order): - When you call some mail-order catalogue companies to place an order, their computers capture your phone number, look you up in the database, and display information about you on the operator's screen even before he or she say hello.

v. When you file a tax return, computers compare information in it against a database from your employer to see if the information matches.

1.4 Database Management Packages

A Database Management System (DBMS) is a set of computer programs that controls the organization, storage, and fast retrieval of data in a database. Examples include: Microsoft, Access, MySQL, Microsoft SQL Server, Oracle, FileMaker Pro, dBase, FoxPro, Sybase.

DBMS can be divided into two categories: -

i. Desktop database ii. Server database

Generally speaking, **Desktop databases** are designed for single user (personal) applications and reside on standard personal computers, hence the term 'Desktop'.

Server databases contain mechanisms to ensure the reliability and consistency of data and are designed for multi-user applications. They are used by organizations to manage big data.

1.5 Components of DBMS

DBMS is made up of the following components: -

a) A modelling Language to define the structure of each database. (e.g. Network, Relational, and hierarchical Model).
b) Data Structures (fields, records, files) to deal with data
c) A dbase query language and report writer to allow for user interaction, (e.g. Structured Query Language (SQL)).
d) A transaction mechanism that ensures data integrity, concurrency control and fault tolerance.

1.6 Functions of DBMS

a) Create records and fields
b) Enter information
c) Add, delete, and alter records and fields
d) Search for information
e) Create reports
f) Sort alphabetically
g) Mail merge (producing personalized letters for a database mail list.
h) Provide security for the database.

1.7 Database Concepts

A database structure consists of a file (table) or set of files (tables) that can be broken down into records (rows), each record is made up of one or more fields (columns). Fields are the basic units a data storage. Users retrieve information primarily through queries. Using keywords and sorting commands. Users can rapidly search, rearrange, group, and select the field in many records to retrieve or create reports on a particular collection of data according to the rules of the DBMS being used.

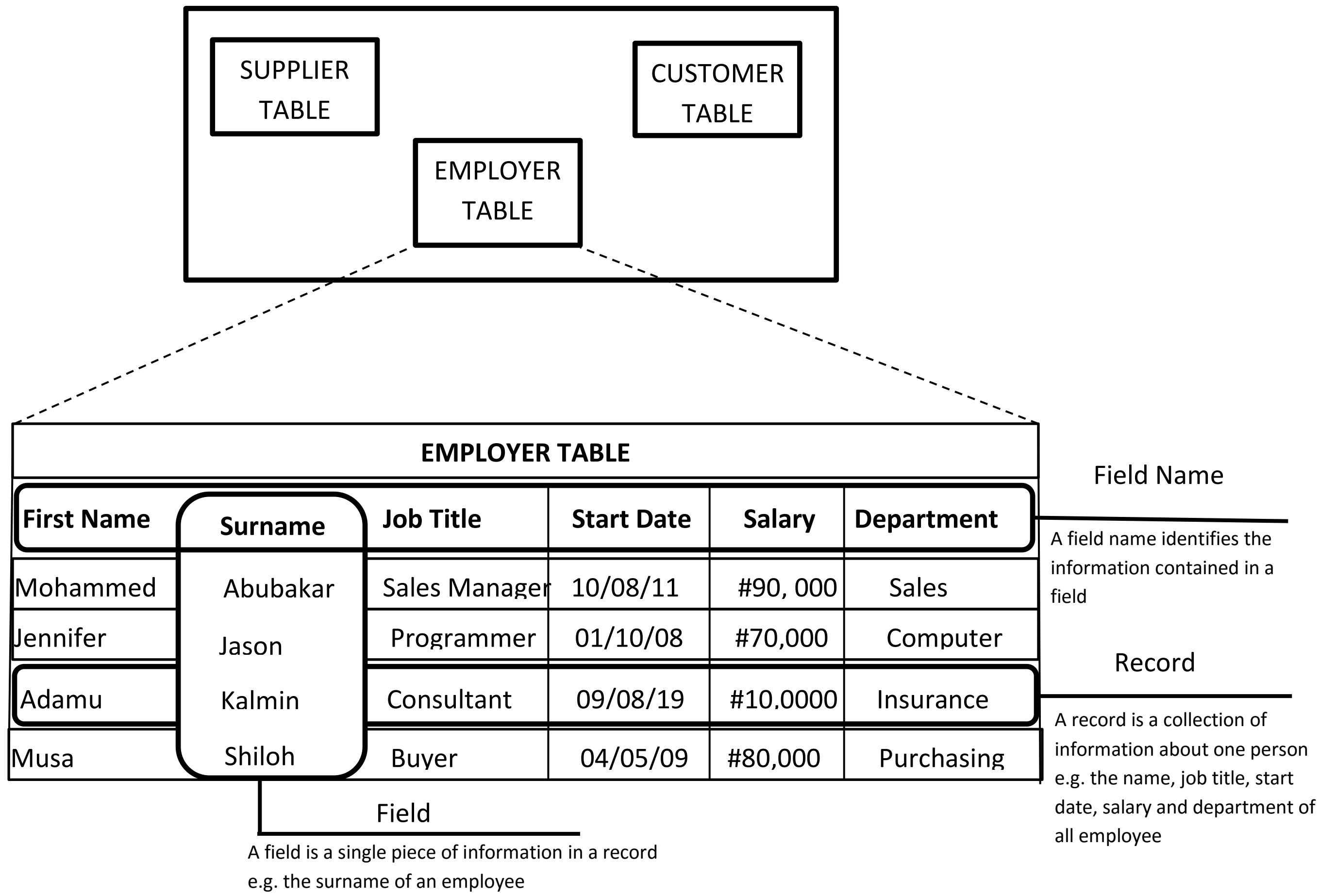


Fig. 1: A company's Database File

FIELD:

A field (column) is a single fact or single piece of information in a record. For example, in an employee's record, things like surname, first name, job title, address, age, salary, etc. are held in separate fields. Each field has a unique name that identifies it. See figure 1 above.

RECORD:

A record or (row) is a collection of fields containing information about a single item in the table. For example, in figure 1, all the details relating to one employee will be held in that employee's record which consist of several fields (columns).

FILE OR (TABLE):

A file (Table) is a collection of records all having the same fields. It contains information about a specific topic, such as employee data in fig. 1.

A database can made up of one or more files (tables). If your database requirements are more complex, your database may contain several tables.

In fig. 1, the company's database has a table for employee data, a table for customer data and a table for product data. The various entries on the employee form for one person are the fields, and all fields for one employee form a record. Finally, all the employee records form a file (i.e. employee table).

1.8 Working with fields

FIELD NAME:

A field name describes or identifies the contents of the field. For example, if a field is to store phone numbers, its name might be *Phone, Phone Number, or Phone #.* When naming fields in a database file, observe the following rules:

- The same field name cannot be used twice in the same table.
- Field names can be up to 255 characters including spaces.
- Don't use period (.), exclamation marks (!), accent grave characters (`), and square brackets ([ ]).

FIELD SIZE:

Afield size property specifies the maximum number of character positions that is set aside for data entry in each field. This number of positions is also referred to as *FIELD WIDTH.* The field size differs for different fields but remains the same for the same field in different records.

For example, each record in the employee table in fig. 1 may be made up of numerous fields, including the following: First Name, Surname, Job Title, Start Date, and Department.

Assuming the following character positions are assigned for the respective fields we will have:

Field Names	First Name	Surname	Job Title	Start Date	Salary	Department
Field Size	8	8	13	8	7	10

Thus, the total length of the record would be 8 + 8 + 13 + 8 + 7 + 10 = 54

You should specify the smallest possible field size setting because the table will then be taken less space and can be processed more quickly. The default width of most data types is set automatically, but you can change Text and Number fields. Text fields can be between O – 25 characters long (the default is 50)

1.9 Data Types

For each field in a database system you must specify data type. This information is used to determine what data type you can enter into the field and how it is displayed and used. For example, if you specify a field's data type as currency, you can enter numbers such as 100.55 in the field but not text such as two hundred. Some of the data types supported by database systems are:

Text fields: - Store names and numbers, such as addresses, phone numbers, admission numbers and postal codes. When numbers are entered into a text field, they are treated as text, not as values. Therefore, they cannot be used in calculations but will retain leading zeros, such as the zero that begins as 0670001 postal code. A text can hold up to 255 characters of data. The default setting is 50 characters.

Memo fields: - store general descriptive text and do not have a fixed length. They expand as you enter up to 64,000 characters of data.

Number fields: - store numbers that can be used in calculations. Entries to these fields can only have numbers, a decimal point, and a plus or minus sign. (To store dollars, naira or, other monetary data, use a currency field).

Data/Time fields: - store dates and times, which can then be used in calculations. For example, you can add or subtract dates or add or subtract numbers to or from them. This allows you to get answers to questions like "What is the average number of days between supplies?" or "How old is someone? Or "is the warranty over 60 days due?"

Currency fields: - Currency fields are used for currency such as Naira, dollars, pounds, or euro. You should use this data type instead of a number data type to store any kind of monetary values.

Counter fields: - Counter fields automatically number records in the order in which they are entered. These numbers uniquely identify each record.

Yes/No fields: - Yes/No fields store only Yes or No (which you can enter as Yes/No, True/False, On/Off).

OLE Object fields: - OLE Object fields store objects created by other applications that support object linking and embedding (OLE). These objects can be graphics, sound, animations, or videos.

1.10 Description

In Microsoft Access DBMS, an optional description for each field is use to provide more information about the field than is provided by its name. A field's description is displayed on the status bar when you select the field in Datasheet view or move the insertion point into its entry space in a form. This is especially helpful when the person 3ntering and maintaining the database is not the same one who created it.

1.11 Primary Keys

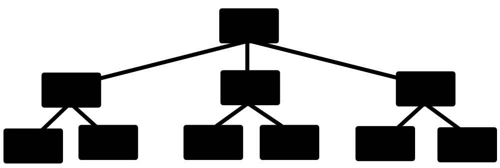
In a database, ideally each table should have a field that uniquely identifies each record. Such a field can be specified as the table's primary key. Tables with primary keys operate faster and can be related to other tables in database that use the same primary key. For a field to be designated as primary keys, each record in the field must contain a unique data. (Microsoft Access will not allow you to store duplicate data in the primary field). Fields containing data such as last names will normally not suffice because if the database gets large enough, there will almost certainly be duplication of names. For example, if a field contains two or more Jason, that data no longer uniquely identifies the field. Many kinds of data can uniquely identify records, including:

- Social security numbers
- Serial numbers
- Driver's license numbers
- Dates and Time (science mainly)
- Employee's serial numbers
- Telephone numbers
- Part number
- Vehicle license number
- Bank account number
- Customer account number
- Purchase order number
- Credit card number

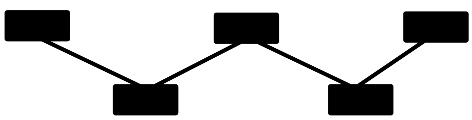
1.12 Types of Database Management System

The physical view of the database involves the use of data models to represent the database.

There are currently three data models that are commonly used for database management systems. These are the Hierarchical data model, the Network data model and the Relational data model. The model tends to determine the query languages that are available ta access the database. One commonly used query language for the relational database is the Structured Query Language (SQL). A query is a statement requesting the retrieval of information, and the set of rules for constructing queries is is known as a Structured Query Language (i.e. SQL). Figurev2 shows a schematic of various data models.



Hierarchical Data Model



Network Data Model

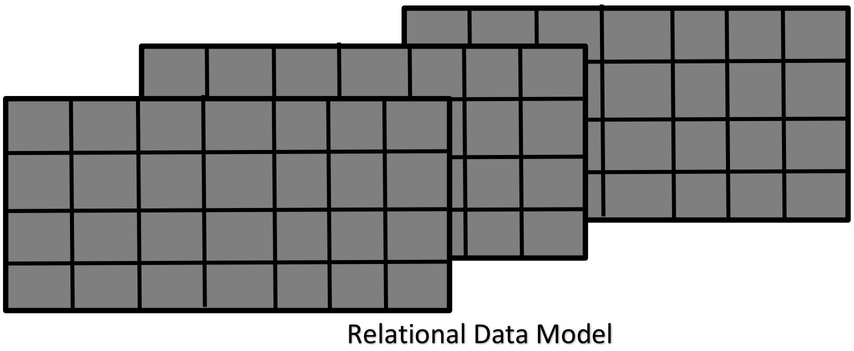


Fig. 2.0: Schematic of various Data Models

Based on these data models, DBMS can be categorized into three types viz:

1) Hierarchical Database Management System.
2) Network Database Management System.
3) Relational Database Management System (RDBMS).

Hierarchical Data Model

In a hierarchical data model, all relationships between data elements are such that there is a hierarchy from top to the bottom with each lower level element linked to only one upper level element. It can be used to handle situations where data elements have an inherently superior – subordinate relationship. Every data element has one and only one parent or owner that may be liked to one or many lower level elements called children. Such linkages are referred to as **one-to-one** and **one-to-many** relationships.

As an example of the use of hierarchical data model, consider the case of a database made up of family names, the cars owned by family members, and the outstanding parking tickets on each car. Figuren3 shows how this automobile database would be represented as a hierarchical data model.

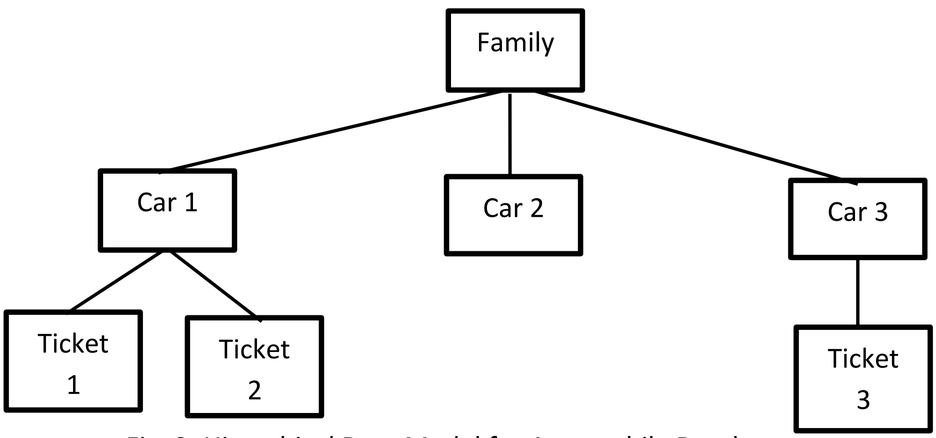


Fig. 3: Hierachical Data Model for Automobile Database

Note in figure 3 that family name is the highest-level data type, the automobiles are the next level, and the parking tickets for each automobile are the bottom. There is always only one upper level data element for any number of lower level elements. Each case is an example of one-to-one or one-to-many relationship.

Network Data Model

There is hierarchy, but a lower level element may be linked to many upper level elements. It is used when there are many-to-many relationships. The network model allows each data element to have more than one parent.

Figure 4 shows an example of how the network model can be used to represent the automobile database when we add another upper level data element – the insurance company that insures each automobile.

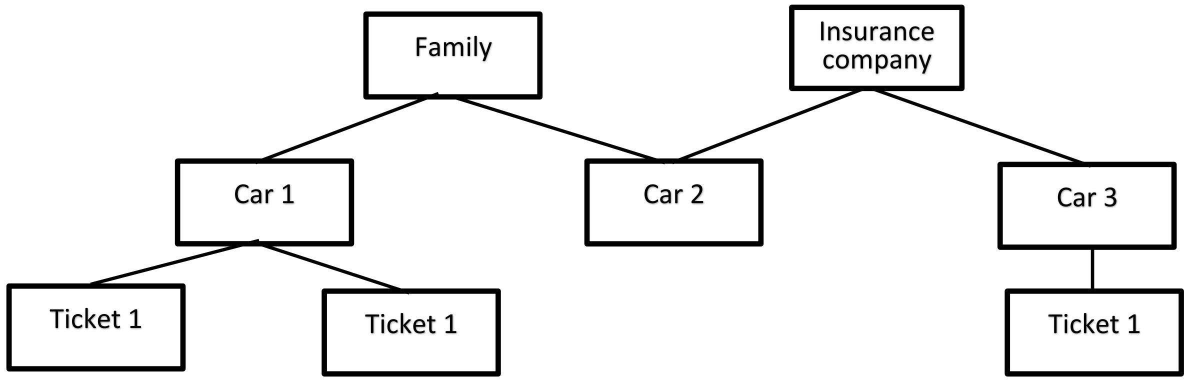


Fig. 4: Network Data Model for Automobile Database

Note: While Hierarchical and Network models use "trees" to show the linkages between fields, the relational data model uses table structures to do this as discussed below.

Relational Database Model

Relational database model stores information in two or more tables. Each table has multiple columns each column has a unique name. each table contain information on a different topic and they are related. If you change the information in one table, the same information will automatically change in all other tables. This makes updating fast and accurate.

Relational database model is powerful and flexible, but difficult to set and learn. This type of database is ideal for invoicing, accounting and inventory. It is the most commonly used database management system.

Figure 5 below is an example of a relational database showing the addresses, owners and outstanding citation tables of vehicle owners with the road safety department. Here by using the vehicle owners License plate number, the road safety violation and the address of that individual can be checked instantly.

Address Table

Name	License	Street Address	City	State
Kalmin Linda	FJ1987	10032 Park Line	Ikeja	Lagos
Umar	D12374	1324 Oak Str.	Mubi	Adamaw
Shiloh M.	LK3457	95874 State Str.	Jos	Plateau
-	-	-	-	-

Key Fields Linked

Key Fields Linked

Owner's Table

Name	Plate Number
Shiloh M.	ABK341
Abram Joy	LMJ198
Jibril	ZXA954
-	-

Outstanding citation Table

License	Citation	Violation
T25475	00031	Speed
D98372	19001	Park
LK3457	69100	Speed
-	-	-

Fig. 5: Relational Database

Review Questions

1. What is a database? What is the Relationship of a data ta a files?
2. What is the database management package? Name three operations oall database management packages perform.
3. What is the difference between a file processing system and a database management system?
4. What are the elements of a file? How are they ordered? What are the field width and field name?
5. List the three data models.which does not support a many-to-many relationship between fields?
6. Why is a relational database package more flexible and powerful than a file processing package? If you were to set up a membership list of a student organization, what type of package will you choose? Why?

1.13 Database Management Software

To create, maintain, and use a database, you use a database management application such as Microsoft Access. Such programs allow you to store information, retrieve it when you need it and update it when necessary.

MICROSOFT ACCESS

Access is a relational database software – this means that all related data is stored in one place. If you are storing data about your business, you could have your employee data, customer data, product data, supplier data, etc all store in your company database. This allows you to compute and analyse quickly and efficiently.

ACCESS OBJECTS:

When you work with access, you create tables to hold your data, queries to find data and forms to make data entry easier, and reports to share with others. Each of these is called an OBJECT. An object is an element you can select and manipulate as a unit.

The database window has a series of tab like buttons down the left side called object buttons. You click these to see the various types of objects that are related to the database. There are seven types of objects in an Access database:

1) **Tables** are where you store the actual data. A database can contain a number of tables, and they can be related to one another. A table consist of records, which contian fields. The information in a table appears in rows (records) and columns (fields).
2) **Queries** are used to get specific data out of the database. You might want to extract records that meet specific selection criteria (e.g. all employees on Grade G in accounts department).
   When you create and then use a query, the results are displayed in a table. However, the query may contain only selected fields and records and it may draw fields from more than one table.
3) **Forms** are used to provide an alternative to tables for entry, display and editing of data in a more attractive and user-friendly format than a table presents.
   You can design your form to look like printed forms that you use. When you use form, you display one record at a time on your screen.
4) **Reports** are used to provide various printed outputs from your database. Using reports on the same database, you can produce a list of customers in a certain area, a set of mailing labels for your letters, or a report on how a customer ows you.

5) **Macros and Modules** are used to automate the way you use Access and can be used to build custom applications.

   Macros can run a series of commands. You can write macros for repetitive activities and to automate the procedures you use most often.

   Modules contain Visual Basic Applications (VBA)mfunctions or procedures. This code can be used to run start-up tasks.

6) **Pages** are hypertext mark-up language (HTML) documents linked to the data. They are dasigned for use in a web browser and requires Internet Explorer.

1.14 Database Construction Using Microsoft access

As mentioned earlier an Access database is a collection of objects, all stored together so they are easy to find and work with. The first step in creating a new database is to assign it a name and specify which directory on the disk it is to saved in. when you have saved the database, then the Database Window is displayed so you can create objects such tables, forms, queries and reports for the new database.

To construct a database the following steps are required: -

1) Create a new Blank Database and it with a file name.
2) Create and Design a New Table.
3) Save the New Table with a Table Name
4) Enter and Edit data in the Table.
5) Save the data in the Table.
6) Preview and print the Table.
7) Close the database.

CHAPTER TWO

COMPUTER AIDED DRAFTING (CAD)

2.0 Introduction

Computer-Aided Drafting (CAD) is the process of using special software and a computer system to produce technical drawings. When design is a major part of the process, the system is sometimes called Computer-aided Drafting and Design (CADD).

CAD has a variety of applications in business and industry. It is used for drawing product plans, such as automobiles, appliances and machinery. Architectural plans are often drawn with CAD. Electronic circuit boards and circuit layouts are drawn with CAD. Even landscaping plans can be drawn in this manner.

Examples of CAD software are AutoCAD and ANSYS 7.0 while examples of CADD software are CATIA V, SolidWorks, NX6, Solid Edge V21, EdgeCam, etc.

2.2 Advantages of CAD

There are several advantages of CAD that are important to consider. These include: -

1) Drawings can be generated more quickly using a CAD system than other means. For example, dimensioning can be done automatically, saving a great deal of drawing time.
2) Editing function, such as erasures, are much simpler done electronically.
3) A design may be modified many times before the best one is chosen. A computer simplifies this task by ollowing for quick modifications.
4) Quality of drawings is better than by using other means. Lines drawn electronically will be consistent in width and density. Text and dimensioning will also be consistent. Human factors, such as the ability to draw a stright, dense line, are not a consideration with CAD systems.
5) Drawings can be stored on magnetic media. Much less space is needed and new drawings can be plotted whenever necessary.
6) Drawings can be transmitted as data to other locations. If two designers are working on the same product in different states, they can easily send their ideas to one another via computer.

2.3 Disadvantages of CAD

1) A computer, software and output devices are needed to draw.
2) People must know ornlearn how to operate the CAD system

2.4 Computer Drafting System

A computer drafting system consists of the hardware and software necessary to create technical drawings. A typical CAD system has a computer system, keyboard and mouse for input, a monitor and plotter or printer can be used as an output device.

2.41 Input Devices

Input devices are used for placing a drawing in the computer. This process is sometimes called **Digitizing.** For example if a point is drawn, this is known as digitizing the point.

The **keyboad** is an input device used with most CAD systems. It can be used for entering text on the drawing. In addition, drawing commands can usually be made using keyboad.

A **Mouse** is one of the most common devices used for input. Movement of the mouse results a corresponding movement of the cursor on the screen. Points are digitized on the screen using the button on mouse.

The **Joystick,** another input device, can be used in a similar manner to the mouse.

The **Graphic Tablet** is another type of input device. It has a flat surface on which the stylus is moved over the grid underneath. A puck or stylus is moved over the surface of the tabley as a locator device. A button is pressed on the puck for digitizing. The stylus digitizes by pressing it against the tablet.

The graphic tablet can also be used to digitize existing drawings by placing the drawing on the tablet and tracing them. Larger digitizing tablets are also used for this purpose. Devices that allow for input of drawings that have already been created are known as **Digitizers.**

2.42 CAD Computer

The computer used for CAD must be compatible with the software. The most sophisticated programs reguire that a hard drive be used in and additional random access memory (RAM). In addition, because of the large number of mathematical calculations needed for computer graphics, a second processor is often needed. This is called a math co-processor.

2.43 CAD Software

CAD software varies considerably in compatiblities and ease of use. Sometimes programs are easy to learn, but are limited in capabilities. Others may be difficult to learn, but have very advanced features. In general, the decision on which software package to use is based on factors such as: -

(1) Cost (2) Ease of use (3) Advanced Features

CAD software produces Vector Graphics, which are also known as Object-Oriented Graphics.

There are several advantages to this type of graphic software:

1) Objects are defined mathematically so they print or plot very precisely, even though they show “jaggies” on the computer monitor.
2) Since drawings are made up of objects instead of dots on the screen, they can be manipulated separately. For example, a circle can be trimmed or moved on the scrssn as a separate entity.

CAD software is usually ***either Menu-driven systems, Command-driven systems, or a combination.*** In MENU-DRIVEN systems, a menu of command options is available on the drawing screen, or a table menu in the case of a graphic tablet. Menus are easy to use, since little memorization of commands is required. A common form of screen menu is the pull-down menu. Whwen an itemis selected from the main menu, other command options appear in column on the screen.

A COMMAND-DRIVEN system use commands typed in from keyboard. This requires either memorization or a reference chart.

Most sophisticated CAD packages provide the conbination option of using the keyboard or menu for entering commands. Experienced users usually prefer this combination since some commande are more quickly entered with the keyboard.

2.44 Output Devices

Output devices are to present the result of CAD. One of the most commmon output devices is the Plotter. A plotter is known as a hardcopy device since a drawing can be permanently stored on a paper. There are a variety of plotters from which to choose. These include the drum plotters, flatbed plotter, and microgrid plotter. Plotters draw with pens. Most plotters provide a choice of pens, so different colours and line widths can be used as necessary.

A pen plotter is used for outputtin technical drawings and other drawings where highquality line work is needed. A pen is used for drawing. A Flatbed plotter is designed so that the substrate remains stationary while the pen moves over the surface horizontally (x-axis) and vertically (y-axis). A Drum plotter uses a cylinder on which the paper moves forward and back. The pen then only needs to move left to right to draw. Another type of plotter, which is similar to the drum plotter, is the Grid-wheel or Microgrid plotter. Instead of a drum, small wheels grip the paper at each side. These wheels turn to move the paper.

2.5 Coordinate Systems

A coordinate system is used in CAD to precisely place or locate entities objects) on a drawing. To specify a point in a plane, take two mutually perpendicular lines as references. The horizontal line is called X-axis, and the vertical line is called the Y-axis. The point of intersection of these two axes is called the Origin. The X and Y axes divide the XY plane into four parts, generally known as quadrants. The X coordinate measures horizontal distance from the origin (how far left or right) on the X axis. The Y coordinate vertical distance from the origin (how for up or down) on the Y axis. The origin has a coordinate values of X=0, Y=0. The origin is taken as the reference for locating any point on the XY plane. The X coordinate is positive if measured to the right of the origin, and negative if measured to the left of the origin. The Y coordinate is positive if measured above the origin, and negative if measured below the origin.

This method of specifying points is called the **Cartesian Coordinate System,** in CAD this is known as **World Coordinate System (WCS)** see figure 2.4. In CAD, the default origin is located at the lower left corner of the graphics area of the screen. The Cartesian coordinate system provides an accurate method for locating points on the screen.

There are three ways that Cartesian coordinates are used to for making drawings. These include:

1) Absolute coordinate
2) Relative coordinate
3) Polar coordinate

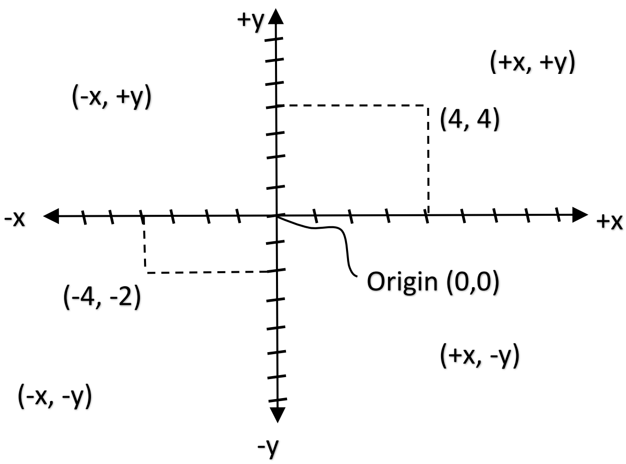


Fig. 2.4

ABSOLUTE COORDINATES

With absolute coordinates, all XY coordinates are located with respect to the origin (0,0), for example, a point with X = 4 and Y = 4 units horizontally and 3 units vertically from the origin, as shown in figure 2.5.

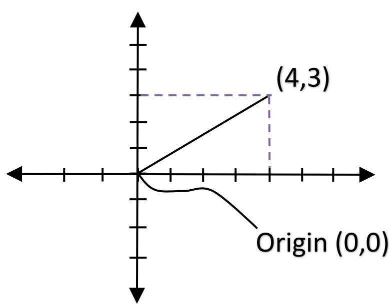


Fig. 2.5

RELATIVE COORDINATES

In the relative coordinate system, the displacements along X and Y axes are measured with reference to the previous point rather than the origin. It is entered as @X,Y. Example, @5,4. This means that a line will be drawn from the 1st point 5 units along X-axis and 4 units along Y-axis, see figure 2.6.

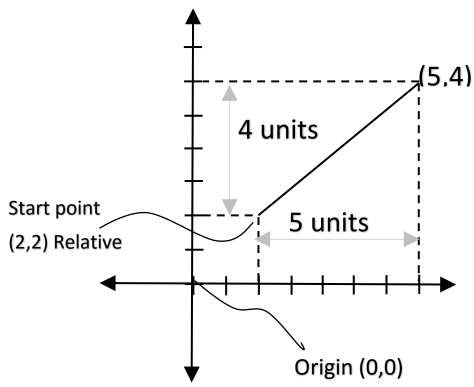


Fig. 2.6

POLAR COORDINATES

In polar coordinate system, a point is located by defining both the distance of the point from the current point and the angle that the line makes with the positive X-axes. This is entered as @D<A. In this case, D is the distance and A is the angle. Example, P (4<35) will give 4 units at an angle 35^0 with the positive x-axis. When drawing lines at an angle, you have to start measuring the angle from 0 degrees (3 o'clock position) and move anticlockwise.

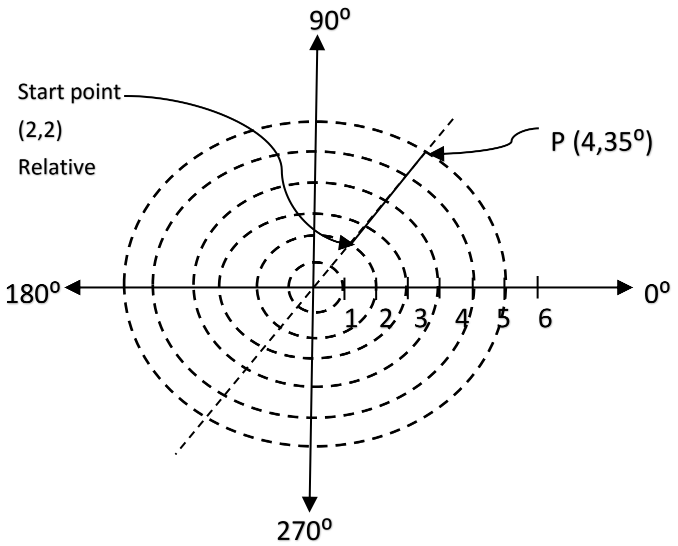


Fig. 2.7

2.6 Unit Measurement

Units are a measure of distance on a drawing. They are the same as the coordinate numbers. For example, X2, Y3 means to move two units to the right and three units up.

The term "units" is used because it can be used to represent different measurement systems. For example, a user might define a unit as 1 inch for a drawing and then 1mm for another drawing. Many CAD systems allow the operator to change the format of measurement. For example, fractional inches could be specified in fractional or decimal form. The coordinate will then be displayed in this same way as the dimensions.

2.7 Entities

Entities are the elements that makeup a drawing. These are sometimes called **primitives.** These include points, lines, arcs, and circles. Entities are combined to make the drawing.

Entities are located on the drawing by using the Cartesian coordinate system. A point can be specified with XY coordinate, such as (2, 4). A line is defined by its end points such as (2, 4) and (3, 6).

2.8 Modifiers

Modifiers further define how any entity is drawn. For example, there are several ways a circle might be drawn using modifiers. In one method, three points can be digitized on the screen and a circle is drawn whose circumference touches the three points. In the other method, the location of the Centre is digitized, and the diameter is entered from the keyboard. In CAD systems, most entities have modifiers that can be used with them.

2.9 Dimensioning

Most CAD systems are capable of automatic dimensioning. However, it is up to the CAD operator to specify where these dimensions are located. This is why CAD user should understand correct dimensioning procedures before dimensioning the drawing. There are several methods that may be used when dimensioning a drawing, depending upon what is being dimensioned. These include: -

1) Linear (horizontal) dimensioning
2) Angular (aligned) dimensioning
3) Diameter dimensioning
4) Radius dimensioning

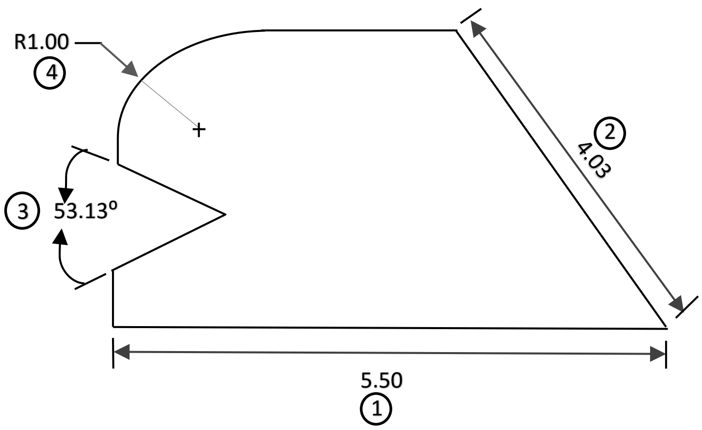


Fig. 2.8

2.10 Drawing Aids

Drawing aids are features of CAD programs that simplify the drawing process. Some of the most common of these include:

1) Grid
2) Grid snap
3) Panning
4) Zoom
5) Layering

GRID: - A grid is a set of reference dots on the screen in rows and columns. The user can specify how far apart these dots should be in units. Entities such as lines can be drawn using the grid points to determine location. The grid is only used as a reference and does not plot out with the drawing.

GRID SNAP: - A function that is often used with the grid is grid snap. Grid snap ensures that what is drawn goes to a point on the grid. This is very useful since it is sometime difficult to be sure that the cursor is directly on a grid dot.

ZOOM: - The zoom command allows the user to enlarge or reduce a portion of the drawing. In some programs, this is known as windowing. Zooming is necessary when working on small details on the drawing that are not clearly shown on the monitor.

Zoom-in refers to enlargement of a portion of a drawing. The amount of enlargement can be specified as part of the zoom command.

Zoom-out or Window-out is used to reduce the drawing size on the screen. In order to fit the entire drawing on the screen, zoom origin, zoom full or window full, are used depending on the program.

PANNING: - Another useful drawing aid is Panning. Panning allows the user to move the drawing on the screen. For example, you may zoom in and find that part of the drawing you wish to work on is half off the screen. Panning is used to centre the magnified view. Panning is accomplished by digitizing at two points on the screen. The second point denotes which direction and how far you wish to move. In some programs, command can be used for this, such as pan left or pan down.

LAYERING: - Layering occurs when drawings are that are meant to overlay one another. Most CAD systems are capable of layering. A useful example of layering is in architectural drawing. A floor plan can be drawn as a first layer. Utilities can then be added to layers, such as electrical on the second layer and plumbing on the third layer. By plotting layers 1 and 2 together, the floor plan is drawn with the electrical details. Layers 1 and 3 provide plumbing details on the floor plan. This technique allows using the floor plan for several drawings without redrawing it. Layers can be plotted separately or together as needed.

2.11 EDITTING

Editing involves changing something that has been previously known. One of the most common editing commands is Erase. Erase allows you to remove one or more entities from the drawing. When several entities are to be removed most CAD programs allows you to create window around those entities to identify the ones to be removed. In some instances, only a part of an entity needs to b removed. In this case the **Trim command** is used. For example, a line or arc can be shortened when this command is used.

The **Move command** allows you to move one or more entities to another location on the drawing.

The **Rotate command** is similar to the move command, but in this case, movement takes place at an angle around a point. The amount of rotation can be specified in several ways, such as digitizing points on the screen or inputting degree of rotation on the keyboard. A group of entities can be by first placing a window around them.

The **Copy command** makes a duplicate of one or more entities that can then be placed elsewhere on the drawing. This is a handy command to be used to avoid redrawing detailed objects, such as screws.

The **Mirror command** copies an entity, or group of entities, in a reverse position from the original, like image. **A Mirror line** is defined and the mirror copy is created on the opposite side of the line. You can draw half of the drawing and then use the mirror command to create the other half copy.

Most CAD systems have additional editing features, but some of the more important commands have been explained here.

2.12 Producing a Drawing by Computer

While steps may vary according to the software package and the type drawing, there is usually a general sequence of events that is followed when producing a drawing by computer. A suggested sequence of events are as follows: -

1) Make a sketch of what is to be drawn, either on grid paper or by computer. This helps in visualizing a drawing and reduces mistakes.
2) Using the necessary procedures, booth the CAD program.
3) As needed set-up parameters for the drawing. These include such aspects as units of measurement, substrate of size for plotting, and drawing scale.
4) To help in creating the drawing, set up a grid on the screen to a convenient size. Use snap grid if needed.
5) Draw entities using absolute, relative or polar coordinates. Use editing commands for corrections and to manipulate the screen. Drawing aids, such as zoom and pan are used when creating detailed portions of the drawing.
6) Dimension the drawing. Use linear, angular radius and diameter dimensioning as needed.
7) Place text on the drawing such as notes.
8) Save the drawing. In fact, you may wish to save the drawing intermittently throughout the process.
9) Plot (print) the drawing.
10) Review, edit and plot the drawing as needed until it is correct.

2.13 CAD/CAM

Using computer to operate machines is known as **Computer-aided Manufacturing (CAM).** These computer-operated machines are known as **Computer Numerical Control (CNC)** machines. Tasks that these tools can perform include drilling, machining, welding and cutting.

In CAD/CAM systems, the CAD system is first used to create drawings of the part to be manufactured. After the drawings of the part to be created, they are stored on magnetic tape or disk that can be used as input to the CNC machine. The parts are then manufactured using the drawings originally created with the CAD system.

2.14 AUTOCAD

Various components of the initial AutoCAD 2009 screen are:

1) **Title Bar:** This will show the program running and what the current filename is.
2) **Menu Bar:** These are standard pull-down menus through which you can access all commands.
3) **Menu Browser:** Used access all menus in the application.

4) **Quick Access Toolbar:** The Quick Access toolbar stores commands that you frequently access. It is a customizable toolbar that contains a set of commands defined by the workspace.
5) **Ribbon:** The ribbon provides a single, compact placement for operations that are relevant to the current workspace.
6) **Drawing Area:** This is where you draw.
7) **WCS Icon:** This is used to show which direction positive X and positive Y go. The WCS means World Coordinate System. (it can be changed to UCS).
8) **Command Window:** The command window is at the bottom of the drawing area. It has the command prompt where commands are entered using the keyboard.
9) **Command Line:** Used to enter commands with the keyboard.
10) **Application Status Bar:** The application status bar displays the coordinate values of the cursor, drawing tools, navigation tools, and tools for Quick View and annotation scaling.

2.15 Drawing and Modifying Commands

To draw using AutoCAD, three basic methods can be used: Keying-in, Icon and the Menu.

The most fundamental object in a drawing is the line. A line can be drawn between any two points using the LINE command. The Line command can be invoked in any of the following ways:

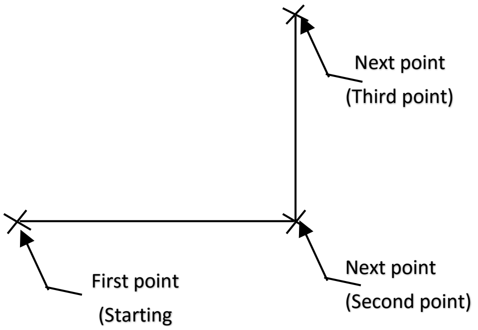
1) Choose the Line button from the Draw panel of the RIBBON. Or
2) Click on the Menu Browser and select Draw > Line.
3) Right click on the Quick Access toolbox and choose Toolbar > AutoCAD > Draw LINE command from the toolbar.
4) Choose View > palettes > Tool Palettes from the Ribbon.
5) Typing LINE or L at the Command prompt.

TO DRAW LINES

STEPS:

1) Command: LINE Enter

2) Specify first point: *Move the cursor (mouse) and left-click to specify the first point.*
3) Specify the next point or [Undo]: *Move the cursor and left-click to specify the second point.*
4) Specify next point or [Undo]: *Specify the third point.*
5) Specify next point or [Close/Undo] Enter (Press ENTER to exit the LINE command)



NOTE:

In case mistakes, type **U** (undo) or Ctrl + Z, or type E (Erase), then press Enter to correct the mistake. The following will take place:

1) After typing U, then ENTER, the previous point will be cancelled. You can also right-click to display the shortcut menu, which gives the **Undo** option.
2) When you press **Ctrl + Z**, the previous point will be automatically cancelled.
3) After typing E, then enter, the cursor will change into an eraser tool. Use it to select the lines you want erased then press ENTER to erase the objects at the same time terminate the ERASE command. Alternately, if you enter A (ALL) at the Select objects prompt, AutoCAD will erase all objects from the screen.